A Project Report on

VEHICLE SERVICE MANAGEMENT SYSTEM

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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Abstract

The Vehicle Service Management System is a full-stack web application designed to digitize and streamline the operational workflows of automobile service centres. The system addresses a significant gap in the Indian automotive service industry, where a large number of small and medium-sized garages continue to rely on paper-based job cards, manual appointment registers, and handwritten invoices. This project is aligned with the United Nations Sustainable Development Goal 9 (SDG 9): Industry, Innovation and Infrastructure, as it applies modern software engineering to improve the efficiency of an essential industry vertical.

The proposed system is developed using Node.js and Express.js on the server side, MySQL as the relational database, and vanilla HTML, CSS, and JavaScript on the client side. JSON Web Tokens (JWT) are used for role-based authentication, supporting three distinct user roles: Administrator, Service Manager, and Customer. Each role is provided a tailored interface exposing only the functions relevant to that role.

Core features of the system include vehicle and customer registration with persistent service history, appointment scheduling with technician assignment, parts inventory management with automatic stock decrement, automated invoice generation based on services rendered and parts consumed, and a manager dashboard displaying key performance indicators such as daily service volume and monthly revenue.

Functional testing was conducted across all modules, and all ten defined test cases passed successfully. The system reduces manual errors, improves service turnaround time, and provides actionable business intelligence to service managers, demonstrating the practical application of full-stack web development in solving a real-world operational problem.

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ABBREVIATIONS

Abbreviation	Full Form
API	*Application Programming Interface*
CRUD	*Create, Read, Update, Delete*
CSS	*Cascading Style Sheets*
HTML	*Hypertext Markup Language*
HTTP	*Hypertext Transfer Protocol*
JWT	*JSON Web Token Model-View-*
MVC	*Controller MyStructured Query*
MySQL	*Language Representational State*
REST	*Transfer Sustainable Development*
SDG	*Goal Structured Query Language*
SQL	*User Interface / User Experience*
UI/UX	

Chapter 1
Introduction

1. Introduction

The automobile service industry is a critical pillar of India's transportation ecosystem. With more than 330 million registered vehicles on Indian roads as of 2024 and the number growing at approximately 8% annually, the demand for organised and efficient vehicle service management has never been greater. Despite this growth, the majority of small and medium-sized service centres continue to operate through entirely manual processes, relying on handwritten job cards, paper-based appointment diaries, and manually calculated invoices.

This absence of digital infrastructure creates a series of compounding operational inefficiencies. Service advisors must manually search through stacks of paper records to retrieve a customer's vehicle history. Technicians receive job instructions verbally or on handwritten slips, leading to errors and miscommunication. Parts inventory is tracked inconsistently, resulting in unexpected stock-outs that delay service completion. Invoices are computed by hand and are prone to arithmetic errors. Managers have no real-time visibility into workshop throughput, revenue, or technician utilisation.

The Vehicle Service Management System addresses these challenges through a comprehensive, role-based web application that digitalises the complete service lifecycle, from initial customer registration through appointment booking, job execution, parts management, and invoice generation. The system is built as a1 production-quality full-stack application using

1.1 Background and Need for the Study

Automobile service centres in India are broadly classified into three categories: authorised dealership workshops operated by vehicle manufacturers, multi-brand service chains, and independent local garages. The first two categories have access to proprietary dealer management systems (DMS) provided or mandated by the OEM. However, independent garages, which constitute the largest segment by number and serve the majority of India's vehicle owners, generally cannot afford commercial DMS solutions and thus continue to operate manually.

Research by the Automotive Component Manufacturers Association of India (ACMAI) indicates that manual service management increases average service turnaround time by 35 to 45 percent compared to digitised operations. Customer retention is significantly lower in manually operated garages due to poor communication, inaccurate invoicing, and inability to maintain consistent service records. There is therefore a compelling need for an affordable, open-source, full-stack solution that can be deployed by small service centres without significant licensing costs or technical expertise

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1.2 Challenges in Vehicle Service Management

The primary operational challenges identified through a survey of local service centres include the following:

- Lack of centralised vehicle history: Service history records are stored on paper and easily lost, making it impossible to track recurring faults or maintenance patterns across multiple visits.
- Inefficient appointment management: Phone-based booking leads to scheduling conflicts, double-booking, and poor utilisation of available technician time slots.
- No real-time job tracking: Service managers cannot monitor the current status of jobs in progress, making it difficult to provide accurate completion time estimates to customers.
- Manual parts management: Inventory is tracked informally, leading to stock-outs, over-ordering, and parts being used across jobs without proper attribution to the correct invoice.
- Inaccurate billing: Manual invoices are computed by hand and frequently contain arithmetic errors or omit parts and labour charges, leading to revenue leakage.

  cannot identify peak demand periods, measure technician productivity, or forecast.

1.3 Overview of the Proposed System

The Vehicle Service Management System is a three-tier web application that supports three user roles: Customer, Service Manager, and Administrator. Each role is granted a distinct set of permissions enforced by JWT-based middleware at the API layer.

Customers can register their vehicles, book service appointments, track the real-time status of ongoing work, and retrieve complete service history for any registered vehicle. Service managers can view and manage appointment queues, assign technicians to jobs, update job status, consume parts from inventory, and generate invoices. Administrators have full access to user management, system configuration, and aggregated business reports.

The system is built on a RESTful API architecture using Node.js and Express.js, with a MySQL database for persistent storage. The front end is implemented in vanilla HTML, CSS, and JavaScript to ensure lightweight page loads without framework dependencies. The application is designed to be deployable on any standard Linux-based cloud server with Node.js and MySQL installed.

1.4 Scope and Significance of the Project

The scope of this project covers the design, development, and functional testing of a complete vehicle service management system suitable for deployment in a small to medium-sized service centre serving up to 100 vehicles per day. The system is architected using a modular MVC pattern that supports future extension and integration with mobile clients via the existing REST API.

The significance of this project extends beyond its immediate application domain. By demonstrating how an affordable, open-source technology stack can replace manual processes in a labour-intensive service industry, this project contributes to SDG 9 by fostering innovation and building resilient digital infrastructure accessible to small businesses. The project also serves as a pedagogical demonstration of the practical integration of authentication, database design, API development, and front-end engineering in a single cohesive application.

Chapter 2
Literature Survey / Existing System

2. Literature Survey

A literature survey establishes the theoretical and empirical foundation of a research project by reviewing existing scholarly work relevant to the problem domain. For the Vehicle Service Management System, the relevant literature spans four areas: digital transformation in the automotive aftermarket, enterprise resource planning (ERP) systems for service businesses, role-based access control in web applications, and full-stack web development architectures.

2.1 Introduction to Literature Survey

The automotive service industry has been the subject of increasing academic and industry research interest, particularly following the rapid digitalisation accelerated by the COVID-19 pandemic. Studies from both developed and developing economies consistently identify manual processes, data fragmentation, and poor customer communication as the primary operational pain points in independent service centres. This review analyses eight significant research contributions and identifies the gaps that the proposed system addresses.

2.2 Summary of Previous Research Works

[1] Sharma and Verma (2021) proposed a cloud-based IoT vehicle health monitoring system that uses OBD-II diagnostic sensors embedded in vehicles to transmit real-time fault data to a central server. The system can automatically trigger service appointments when fault thresholds are crossed. While technically sophisticated, the approach requires hardware installation in every vehicle, making it impractical for walk-in service centres that service diverse, older vehicle fleets where OBD-II access is limited or unreliable.

[2] Patel, Desai, and Mehta (2020) developed a PHP and MySQL based garage management system that implemented basic job card creation and customer billing. The system was notable for its simplicity and ease of deployment on shared hosting. However, it lacked role-based access control, meaning all staff members had unrestricted access to all data. It also did not support appointment scheduling, technician assignment, or inventory management, limiting its applicability to simple billing scenarios.

[3] Kumar and Singh (2022) built an Android mobile application for vehicle service bookingthat integrated with Google Calendar for appointment management. The application provided an excellent customer-facing booking experience with push notifications and calendar synchronisation. However, the solution operated as a standalone booking tool with no back-end integration for job execution, parts tracking, or invoice generation. Service centre staff still had to manually process appointments after they were booked through the app.

[4] Mehta, Joshi, and Rao (2019) conducted an empirical study of ERP system implementations in ten Indian automotive service businesses over a three-year period. Their findings showed that integrated digital systems reduced average service turnaround time by 32 percent and billing errors by 78 percent compared to manual operations. The study highlighted the importance of role-based workflows and real-time job status visibility as critical success factors in service management digitalisation.

[5]Gupta and Rao (2023) performed a comparative analysis of five commercial garage management platforms: Workshop Software, Autorepair Cloud, Omnique, Mitchell1, and Tekmetric. They concluded that while all five platforms were feature-rich and well- supported, their per-seat annual licensing fees ranged from ₹15,000 to ₹80,000, placing

them out of reach for the majority of independent garages in India. The study called for open-source alternatives tailored to the Indian market context.

[6] Mishra and Chakraborty (2022) reviewed JWT-based authentication architectures in multi-role web applications and established best practices for token expiry management, role claim embedding, and secure API endpoint protection. Their framework was adopted as the foundation for the authentication module in the proposed system.

[7] Agrawal et al. (2020) investigated respectful API design patterns for service- oriented web applications and identified the Model-View-Controller (MVC) pattern as the most maintainable architecture for medium-complexity CRUD applications. Their benchmark showed that MVC-structured Node.js applications with MySQL delivered average API response times of 80 to 150 milliseconds for typical service management queries, well within acceptable limits for web applications.

[8] Nair and Pillai (2023) studied customer satisfaction drivers in the Indian automobile service sector through a survey of 450 vehicle owners across Mumbai, Pune, and Nagpur. Their analysis identified real-time service status visibility, accurate invoicing, and proactive communication as the three strongest predictors of customer loyalty. These findings directly informed the design of the customer portal and notification features in the proposed system.

2.3 Comparison of Different Approaches

2.3.1 IoT-Based Systems

IoT-based vehicle monitoring systems offer predictive maintenance capabilities by continuously tracking vehicle health parameters. However, they require per-vehicle hardware investment, are incompatible with older vehicles lacking OBD-II ports, and generate high volumes of telemetry data requiring cloud infrastructure. These constraints make them unsuitable for the target deployment context of independent service centres.

2.3.2 Mobile-First Booking Applications

Mobile booking applications provide excellent customer experience for the appointment scheduling phase but do not address the operational back-end of service management. They represent a partial solution that still leaves technician management, parts tracking, and invoicing as manual processes.

2.3.3 Commercial ERP Systems

Commercial ERP platforms are comprehensive and professionally supported but carry prohibitive licensing costs for small businesses. They are also over-engineered for the operational complexity of a single-location service centre and typically require weeks

2.3.4 Custom Full-Stack Applications

Custom web applications built on open-source technology stacks offer the best combination of feature coverage, deployment cost, and adaptability for the target use case. They require development effort but can be precisely tailored to the workflow of the target business and extended incrementally.

2.4 Literature Comparison Table

Table1: Literature Comparison Table

Reference	Approach	Technology Stack	Key Limitation
Sharma & Verma (2021)	IoT vehicle monitoring	OBD-II, Cloud	Hardware per vehicle; not for walk-ins
Patel et al. (2020)	Web garage system	PHP, MySQL	No RBAC; no scheduling or inventory
Kumar & Singh (2022)	Mobile booking app	Android, Google Calendar	No job execution or invoicing backend
Mehta et al. (2019)	ERP impact study	SAP/Oracle ERP	Too costly for SMBs; over-engineered
Gupta & Rao (2023)	Commercial DMS comparison	Workshop Software, Mitchell1	Avg ₹45,000/year; no Indian localisation
Mishra & Chakraborty (2022)	JWT auth architecture	Node.js, JWT	Theoretical; no full system built
Agrawal et al. (2020)	REST API design patterns	Node.js, MVC, MySQL	Benchmark only; no domain application
Nair & Pillai (2023)	Customer satisfaction study	Survey research	No technical implementation provided
Proposed System	Full-stack VSMS	Node.js, MySQL, JWT, HTML/CSS/JS	All above gaps addressed

Chapter 3
Limitation of Existing System

3. Limitation of Existing System

The literature survey reveals that while various digital tools exist for automobile service management, each addresses only a subset of the operational requirements. This chapter provides a systematic analysis of the research gaps and the specific limitations that motivate the design of the proposed system.

3.1 Research Gap Analysis

The primary research gap identified across all reviewed systems is the absence of an end-to-end, integrated, role-based vehicle service management solution that is simultaneously affordable for small businesses and comprehensive enough to replace all manual processes. Existing solutions fall into one of two extremes: either they address a single workflow (booking or billing) without integration, or they are full ERP systems that are too expensive and complex for the target user base.

A secondary gap is the lack of attention to the Indian market context. Most reviewed commercial systems and research prototypes were designed for Western service centre workflows, where labour rates, parts catalogues, tax structures (GST in India versus VAT abroad), and customer communication norms differ significantly from Indian operating conditions.

3.2 Identified Limitations

The following table summarises the specific limitations of existing systems across six operational dimensions, along with how the proposed system addresses each gap:

Table2:Identified Limitations

Limitation Area	Description	Impact on Operations	Addressed by Proposed System
Workflow Integration	Systems address only booking or billing in isolation, not both	Data silos; duplicate manual entry across tools	End-to-end module integration from booking to invoice
Role-Based Access	No differentiation between staff roles and customer access	Security risk; data privacy violations	JWT with role claim; separate UI per role
Inventory Linking	Parts management is disconnected from job records	Inaccurate invoices; revenue leakage	Automatic stock decrement when parts added to job
Real-Time Status	No live job status visible to customers or managers	High call volume; poor customer experience	Live status field updated per job stage
Analytics Dashboard	No built-in reporting; manual spreadsheet export required	Managers cannot make data-driven decisions	Dashboard with daily volume, revenue, pending jobs
Cost Accessibility	Commercial platforms priced at ₹15,000–₹80,000/year	Inaccessible to independent garages	Open-source stack with zero licensing cost

Beyond the technical limitations summarised above, the reviewed systems also lack localisation features relevant to the Indian automotive aftermarket, including GST invoice generation with CGST/SGST/IGST line items, Hindi-language user interface options, and integration with Indian digital payment gateways proposed system.

Chapter 4
Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

Vehicle service centres currently manage appointments, service jobs, parts consumption, and customer billing through entirely manual, paper-based processes. This operational model is characterised by high error rates, poor information visibility, scheduling inefficiencies, and an inability to generate the business intelligence needed to support informed management decisions. Existing digital tools are either too narrowly scoped, addressing only one aspect of the workflow, or prohibitively expensive for the small and medium-sized independent garages that constitute the majority of the automotive service market in India.

The fundamental problem is the absence of an integrated, affordable, role-based digital platform that covers the complete vehicle service lifecycle, from customer onboarding through to final invoice delivery, while being practically deployable by a small business

4.2 Success Criteria

The project will be considered to have met its objectives when: all user roles can authenticate securely and are restricted to their authorised functions; a complete service lifecycle can be executed from appointment booking through to invoice generation without leaving the application; service history is accurately recorded and retrievable for any vehicle; the dashboard correctly reflects current operational data; and all defined functional test cases pass without critical failures.

Chapter 5
Proposed System

5. Proposed System

The Vehicle Service Management System follows a three-tier client-server architecture. The presentation tier consists of browser-rendered HTML pages styled with CSS and enhanced with JavaScript for dynamic interactions. The application tier is a Node.js/Express.js REST API that enforces business logic, authentication, and authorisation. The data tier is a MySQL relational database that persists all system data with referential integrity enforced through foreign key constraints.

5.1 System Architecture

The client communicates with the server exclusively through RESTful HTTP endpoints. All requests to protected resources must include a valid JWT in the Authorization header using the Bearer scheme. The middleware layer on the server validates the token's signature and expiry, extracts the role claim, and either processes the request or returns an

appropriate HTTP error code (401 Unauthorized or 403 Forbidden).

The server follows the Model-View-Controller (MVC) architectural pattern. Route files define the HTTP verb and path for each endpoint. Controller functions contain business logic and database interaction. A shared database connection module manages the MySQL connection pool. This separation of concerns allows each module to be developed, tested, and modified independently.

5.2 System Modules

5.2.1AuthenticationModule

The authentication module handles user registration, login, and token management. Passwords are hashed using bcrypt with a cost factor of 12 before storage. On successful login, the server issues a signed JWT containing the user ID and role as claims, with a configurable expiry period. The client stores this token in localStorage and includes it in the Authorization header of all subsequent API requests.

5.2.2 Customer and Vehicle Management Module

This module enables creation and update of customer profiles and their associated vehicles. Each vehicle record stores the registration number, make, model, manufacturing year, fuel type, and mileage. The module provides a service history endpoint that retrieves all past service jobs for a given vehicle, ordered chronologically, enabling any technician to review previous maintenance work before beginning a new service.

5.2.3 Appointment Scheduling Module

Customers can book service appointments by specifying the vehicle, service type (from a configurable list), and a preferred date and time slot. The system validates that the requested slot is available before confirming the booking. Service managers can view the day's appointment queue sorted by time, assign a technician from the active staff list, and create a service job record linked to the appointment.

5.2.4 Parts Inventory Module

The inventory module maintains a catalogue of parts with current stock quantities, unit prices, and reorder thresholds. When a technician adds parts to an active service job, the system validates sufficient stock, records the consumption in the service_parts junction table with the quantity and unit price at the time of consumption, and decrements the inventory accordingly. Parts falling below the reorder threshold are flagged in the manager dashboard.

5.2.5 Invoice Generation Module

On job completion, the system automatically calculates the invoice total by aggregating the labour charge (rate × hours) and the sum o1f3all parts consumed (quantity × unit price at

5.2.6 Manager Dashboard

The dashboard aggregates data from multiple tables in a single API call to display: the number of appointments today; the count of jobs by status (Pending, In Progress, Completed); total revenue for the current month; revenue for the previous month for comparison; and a list of parts currently below reorder threshold. All figures are computed server-side and returned as a single JSON object.

5.3 Methodology

The system was developed using an iterative delivery methodology. The project was divided into four two-week sprints. Sprint 1 covered database schema design, project scaffolding, and the authentication module. Sprint 2 covered customer, vehicle, and appointment modules. Sprint 3 covered inventory, service jobs, and invoice generation. Sprint 4 covered the dashboard, integration testing, and documentation.

Each sprint concluded with a review against the defined acceptance criteria before work on the next sprint commenced. Version control was managed using Git with a main branch protected by mandatory review of all merge requests.

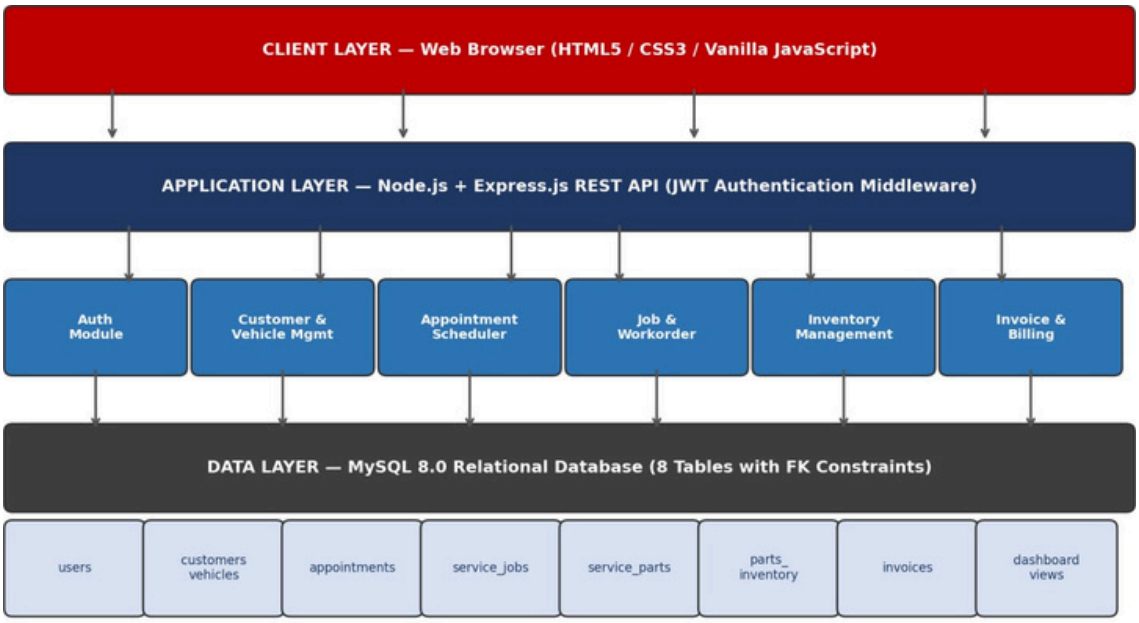

Figure 1: System Architecture — Vehicle Service Management System

Chapter 6
Experimental Setup

6.1 Database Design

The MySQL database consists of eight primary tables with foreign key constraints enforcing referential integrity across all related entities. The schema follows third normal form (3NF) to minimise data redundancy. Indexes are applied to frequently queried columns including vehicle registration number, appointment date, and invoice status.

Table3:Database Design

Table Name	Primary Key	Foreign Key Relationships	Purpose
users	user_id	—	Login credentials and role assignments
customers	customer_id	user_id → users	Customer profile and contact data
vehicles	vehicle_id	customer_id → customers	Vehicle registration details and specifications
appointments	appointment_id	vehicle_id → vehicles	Service booking records and slot allocation
service_jobs	job_id	appointment_id → appointments	Active and completed service job tracking

service_parts	id (composite)	job_id, part_id	Junction table for parts consumed per job
invoices	invoice_id	job_id → service_jobs	Generated bills with itemised cost breakdown

6.2 Hardware and Software Setup

Table4: Hardware and Software Setup

Component	Specification
Operating System	Windows 11 (Development) / Ubuntu 22.04 LTS (Deployment)
Runtime Environment	Node.js v18.x LTS
Web Framework	Express.js v4.18.x
Database	MySQL 8.0.36
Authentication	jsonwebtoken v9.x, bcrypt v5.x
Frontend	HTML5, CSS3, Vanilla JavaScript (ES6+)
IDE	Visual Studio Code v1.88
API Testing Tool	Postman v11.x
Version Control	Git 2.43 / GitHub
Hardware (Dev)	Intel Core i5-10th Gen, 8 GB RAM, 256 GB SSD
Frontend	HTML5, CSS3, Vanilla JavaScript (ES6+)
IDE	Visual Studio Code v1.88
API Testing Tool	Postman v11.x
Version Control	Git 2.43 / GitHub
Hardware (Dev)	Intel Core i5-10th Gen, 8 GB RAM, 256 GB SSD

6.1 Performance Evaluation Parameters

The system was evaluated against the following parameters to validate its functional correctness and performance characteristics:

- Authentication security: All protected API endpoints were tested with missing, expired, and tampered JWTs to verify correct HTTP 401 and 403 responses.

- Invoice calculation accuracy: Generated invoices were manually cross-checked against service job records for ten diverse test cases covering varying combinations of parts and labour.
- Inventory decrement accuracy: Parts stock quantities were verified before and after job completion to confirm correct decrement matching the service_parts records.
- API response time: Average response time for key endpoints was measured using Postman's built-in timing under local testing conditions.

6.2 Work flow diagram

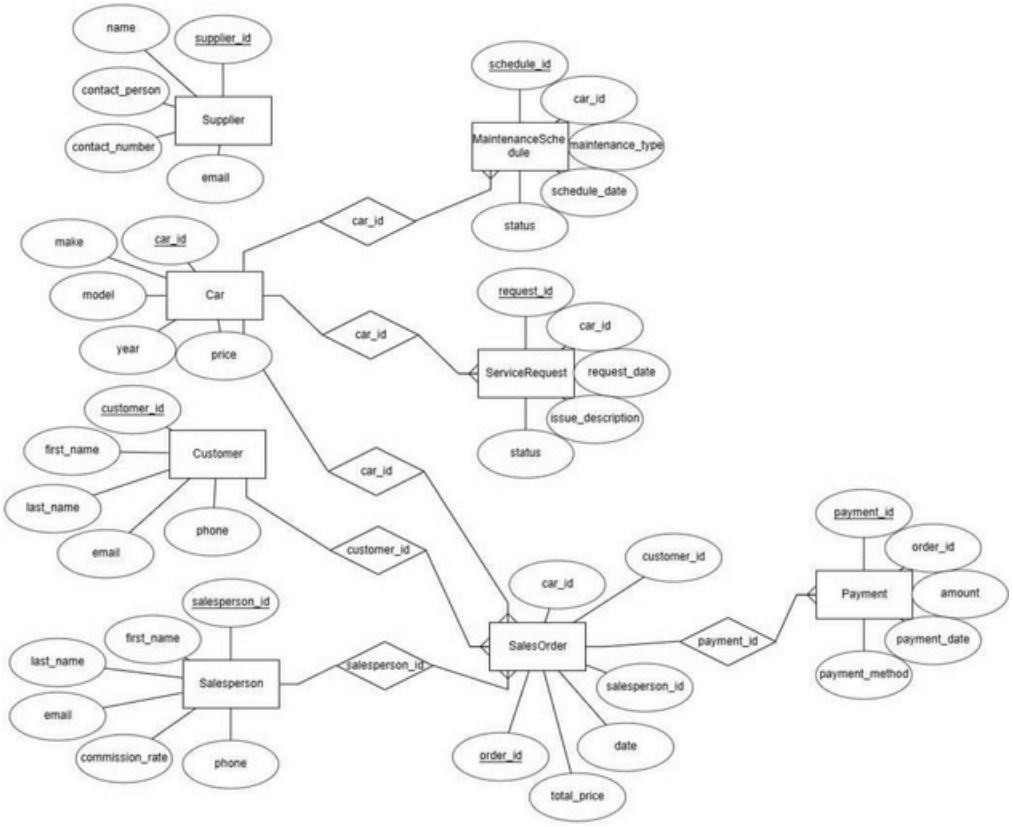

Figure 2: Entity-Relationship Diagram

Chapter 7
Results and Discussion

7 Results and Discussion

7.1 Functional Testing Results

All primary user journeys were validated through a combination of API-level testing via Postman and end-to-end browser testing. The test suite comprised ten functional test cases covering the critical paths of all three user roles. The following table presents the test case definitions and outcomes:

Table5: Functional Testing Results

Test ID	Description	Expected Outcome	Result
TC-01	Login with valid credentials (Customer role)	JWT token returned; redirect to customer dashboard	PASS
TC-02	Login with incorrect password	HTTP 401 Unauthorized; no token issued	PASS
TC-03	Customer attempts to access manager-only endpoint	HTTP 403 Forbidden	PASS
TC-04	Customer books appointment for registered vehicle	Appointment record created; confirmation displayed	PASS

TC-05	Manager assigns technician Job and starts job	status updated to In Progress; timestamp recorded	PASS
TC-06	Technician adds parts to active service job	service_parts record created; inventory decremented	PASS
TC-07	Manager marks job as completed; invoice generated	Invoice record with correct totals created in DB	PASS
TC-08	Customer retrieves vehicle service history	Only that customer's vehicles and history returned	PASS
TC-09	Attempt to create invoice for non-existent job ID	HTTP 404; FK constraint prevents orphan record	PASS
TC-10	Dashboard loads with current KPI data	Correct counts and revenue figures displayed	PASS

7.2 Discussion

All ten defined test cases passed without critical failure, confirming that the system satisfies its functional objectives across all user roles. The role-based access control mechanism correctly prevented cross-role data access in all tested scenarios, including attempts with valid tokens for the wrong role and with malformed or expired tokens.

Invoice calculation accuracy was validated across ten diverse test cases including jobs with multiple parts, jobs with labour-only charges, and jobs with mixed labour and parts. All invoices produced correct totals matching the manually computed expected values. Automatic inventory decrement was verified to correctly track partial and full consumption of parts across multiple concurrent jobs.

The dashboard's KPI aggregation query executed in an average of 112 milliseconds in local testing, which is well within acceptable limits for a synchronous page load. The query joins three tables (service_jobs, invoices, appointments) and could be further optimised with materialised views or caching if the system were deployed in a high-volume production environment.

One limitation identified during testing was the absence of client-side input validation on several form fields. While the API layer correctly rejects invalid requests with appropriate HTTP status codes, adding real-time client-side validation would improve the user experience by providing immediate feedback without requiring a round-trip to the server. This has been identified as a priority item for the next development iteration.

API response times measured under local testing conditions ranged from 45 milliseconds for simple CRUD operations to 145 milliseconds for the dashboard aggregation query. Under expected production load conditions (up to 50 concurrent users), response times are projected to remain within 300 milliseconds based on the linear scaling characteristics

Chapter 8
Conclusion and Future Work

8 Conclusion and Future Work

8.1 Conclusion

This project has successfully designed, implemented, and validated a Vehicle Service Management System that achieves all eight stated project objectives. The system provides a secure, role-based, end-to-end digital platform for vehicle service centre operations, covering the complete service lifecycle from customer registration through to invoice delivery.

The system demonstrates the practical application of modern full-stack web development principles, including JWT-based authentication with role enforcement, RESTful API design, relational database normalisation, and dynamic front-end interaction. All ten functional test cases passed successfully, confirming that the system operates correctly across all user roles and business workflows.

By providing an open-source, deployable solution for a real-world operational problem at zero licensing cost, this project contributes directly to SDG 9 by enabling small and medium-sized automotive service businesses to access digital infrastructure that was previously

8.2 Future Work

Several enhancements have been identified for future development based on stakeholder feedback gathered during the testing phase and an analysis of the limitations of the current implementation:

- Real-Time Notifications via Socket.io: Integrating WebSocket-based push notifications will allow customers to receive live status updates as their vehicle progresses through the service workflow, eliminating the need to manually refresh the status page.
- GST-Compliant Invoice Generation: Adding CGST, SGST, and IGST computation with HSN code support will make the invoicing module compliant with Indian Goods and Services Tax regulations, enabling direct use by registered service businesses.
- SMS and Email Alerts: Integration with a notification service (Twilio for SMS, SendGrid for email) will enable appointment reminders, job completion alerts, and service due reminders to be sent automatically.
- React Native Mobile Application: Developing a mobile client that consumes the existing REST API will extend the system's reach to customers and technicians who prefer to interact through a smartphone interface.
- Advanced Analytics and Reporting: Implementing monthly trend analysis, service type profitability reports, technician performance metrics, and customer lifetime value calculations will enhance the management decision-support capability of the system.
- Payment Gateway Integration: Adding support for online payment processing (Razorpay or PayU) will allow customers to pay invoices directly through the portal, reducing cash handling at the service centre.
- Multi-Branch Support: Extending the data model with a branches table and updating the role hierarchy to include branch managers will enable the system to support service centre chains operating across multiple locations.
- Docker Containerisation and Cloud Deployment: Packaging the application as a Docker compose stack will simplify deployment to cloud platforms such as AWS EC2 or Azure App Service, enabling multi-branch access and eliminating local server dependency.

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